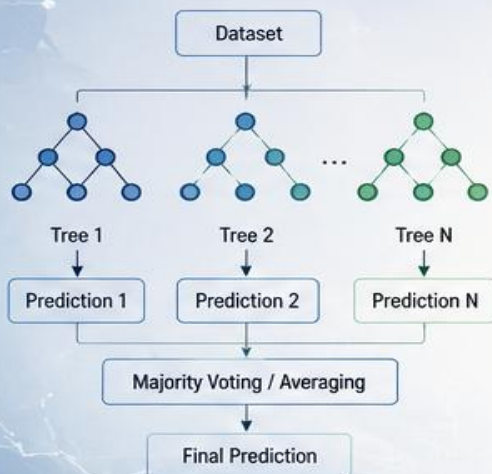


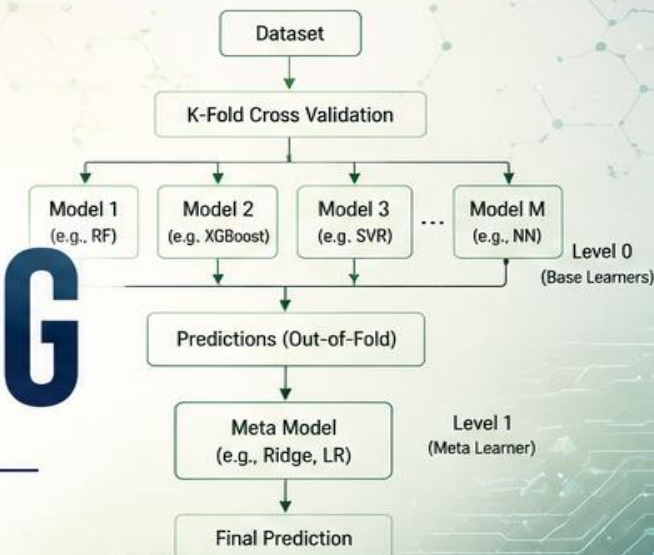


APPLICATIONS OF MACHINE LEARNING IN LITHIUM-ION BATTERIES

RANDOM FOREST



STACKING ENSEMBLE LEARNING



SOC

State of Charge



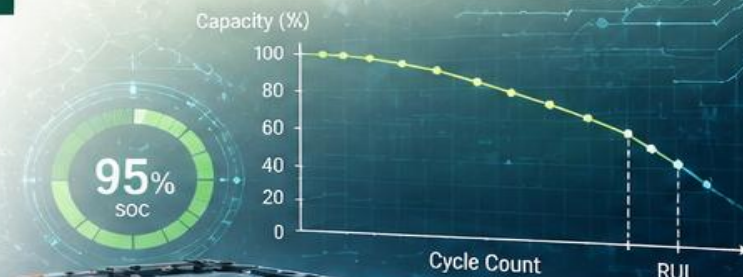
SOH

State of Health



RUL

Remaining Useful Life



PRESENTED BY:

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Kian Mamdoohi**



COURSE INSTRUCTOR:

Dr Ebrahimpour



PRESENTATION DATE:

20 June



Introduction



SOC

①



SOH



RUL

②



Introduction



SOC

①

① State Of Charge



Air Pollution:
A Global Challenge!



① State Of Charge

Transportation Sector Contribution to Global CO₂ Emissions (According to the IEA) $\xrightarrow{2020}$ 24%



① State Of Charge



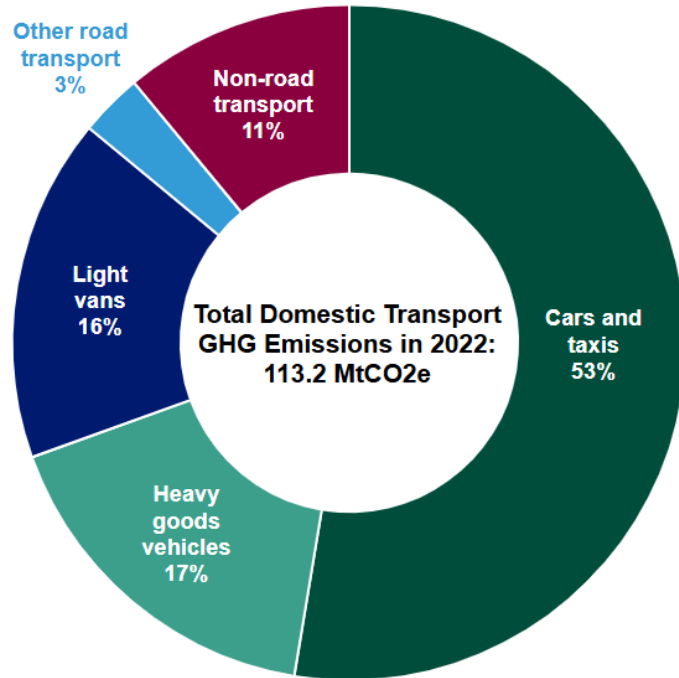
Transportation Sector Contribution to Global CO₂ Emissions (According to the IEA) — 2020 → 24% — 2050 → 50%





① State Of Charge

Transportation Sector Contribution to Global CO₂ Emissions (According to the IEA) $\xrightarrow{2020}$ 24% $\xrightarrow{2050}$ 50%

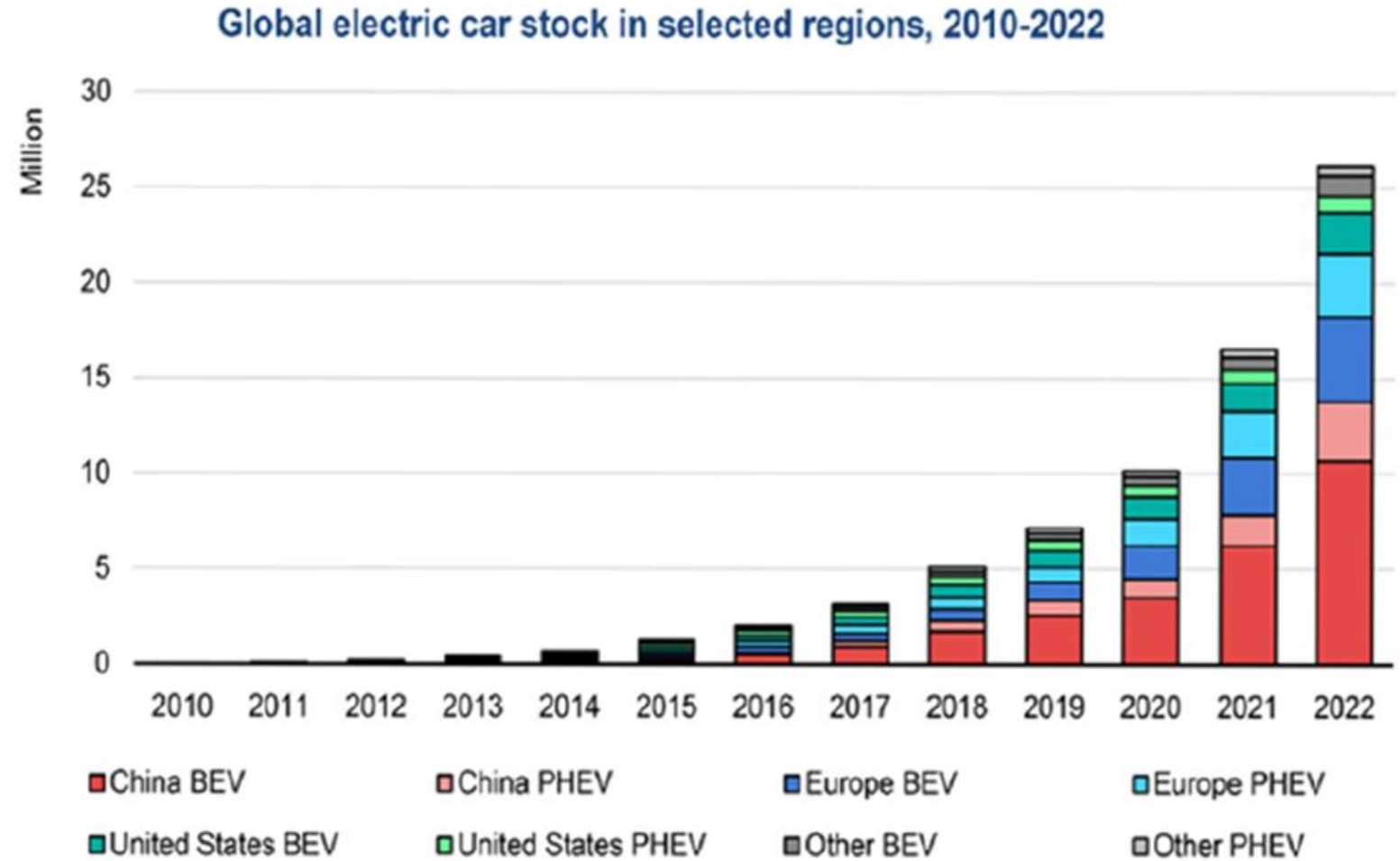


① State Of Charge

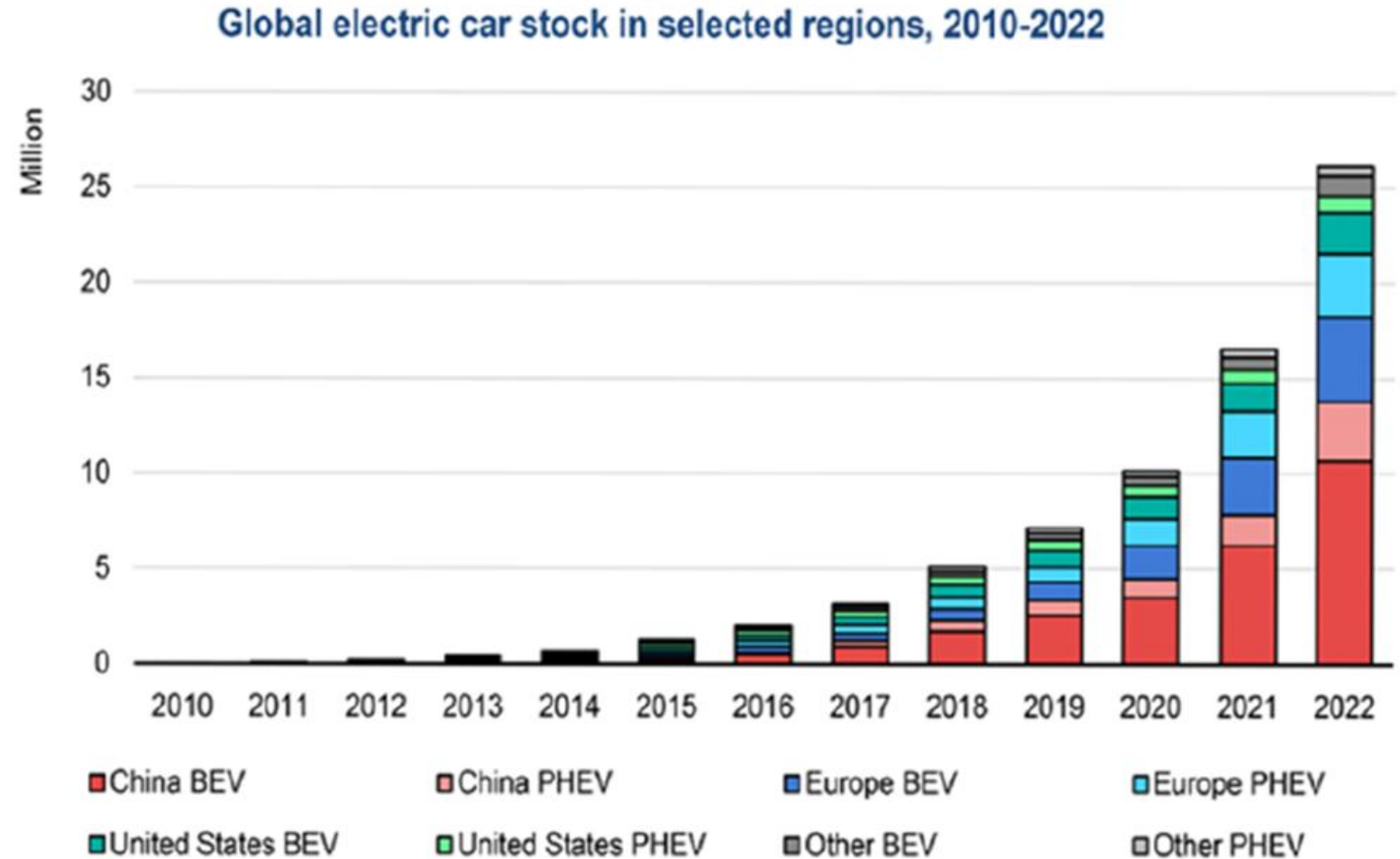
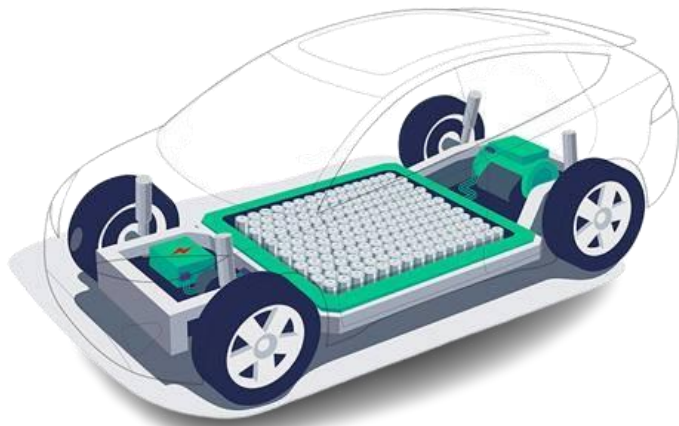
Electric Vehicles



① State Of Charge

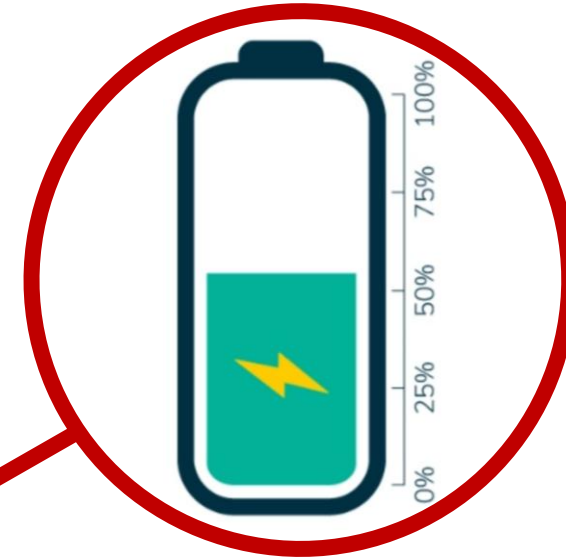
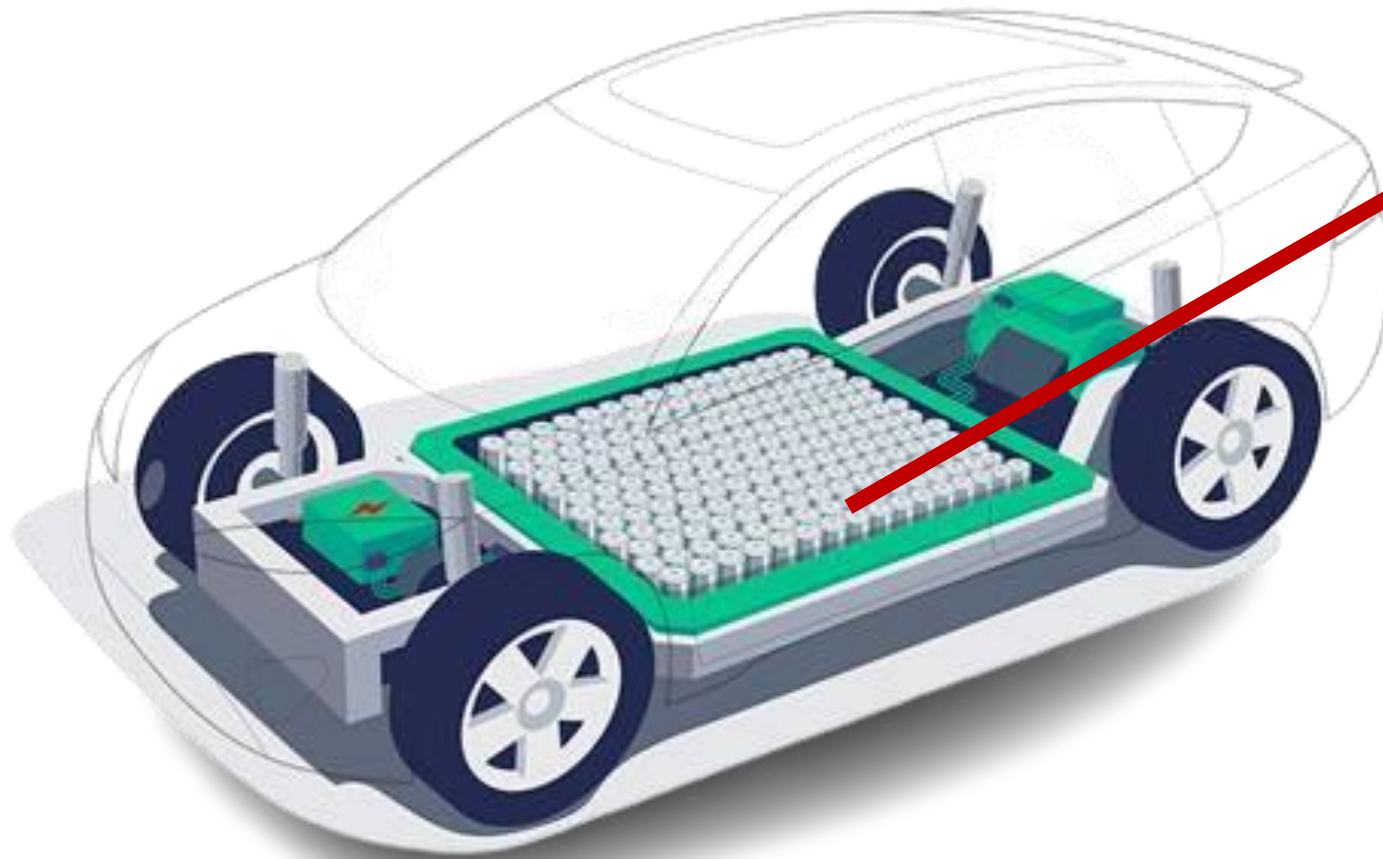


① State Of Charge



① State Of Charge

Rechargeable Batteries Lithium-Ion Batteries



① State Of Charge



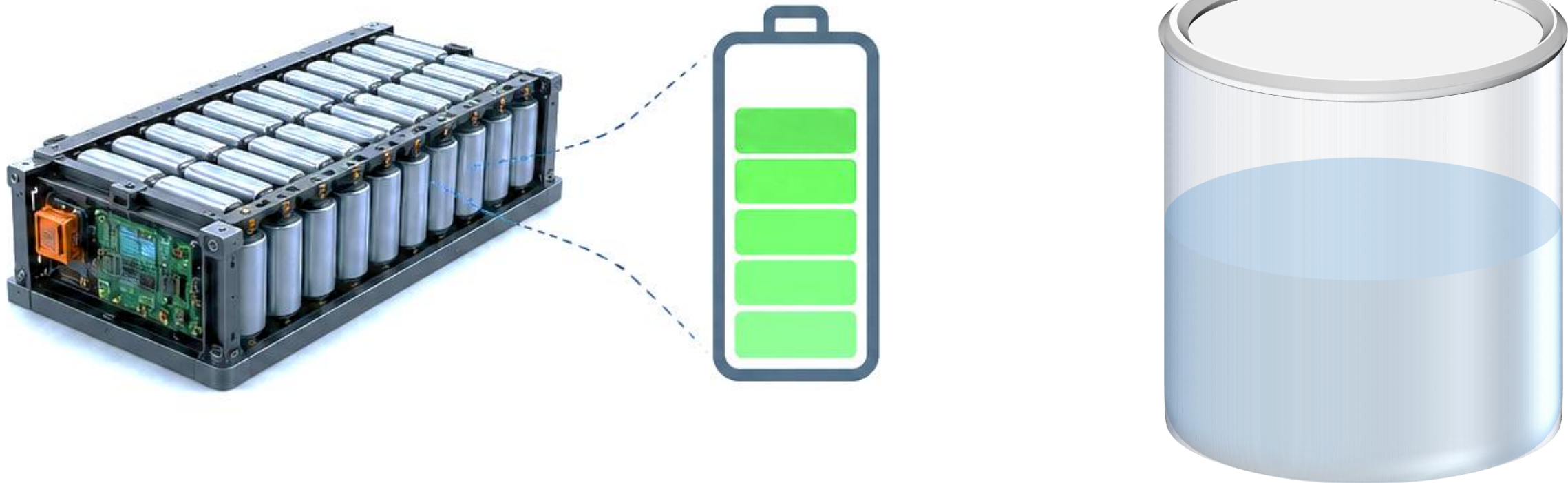
State Of Charge (SOC): The percentage of energy remaining in a battery relative to its full capacity.



① State Of Charge



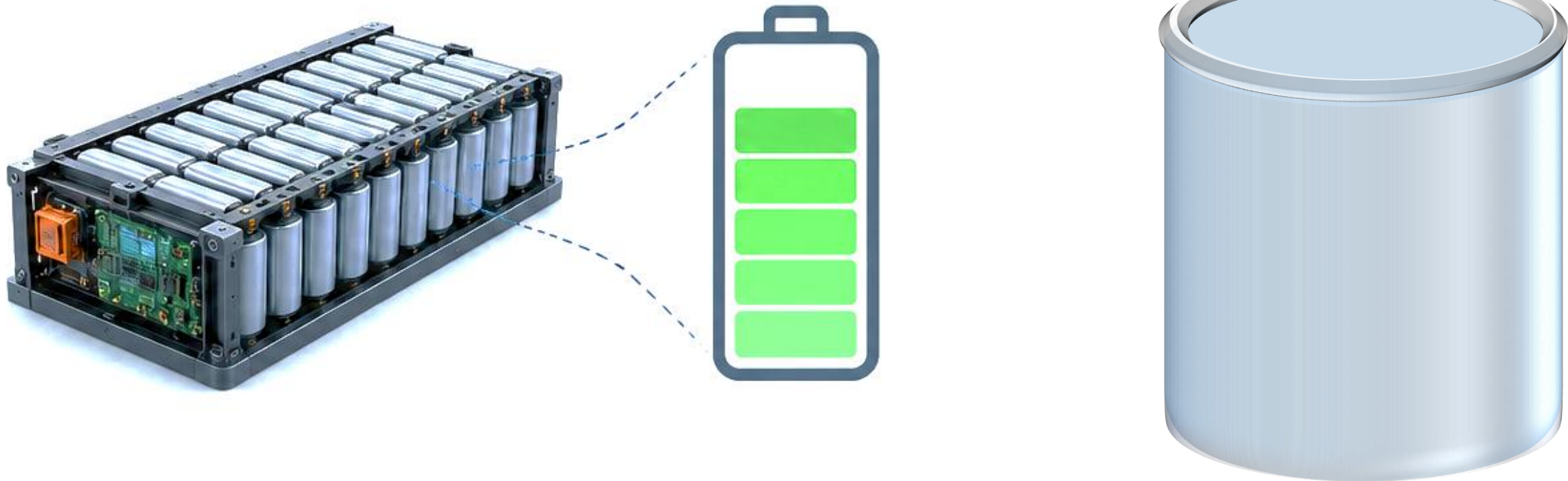
State Of Charge (SOC): The percentage of energy remaining in a battery relative to its full capacity.



① State Of Charge



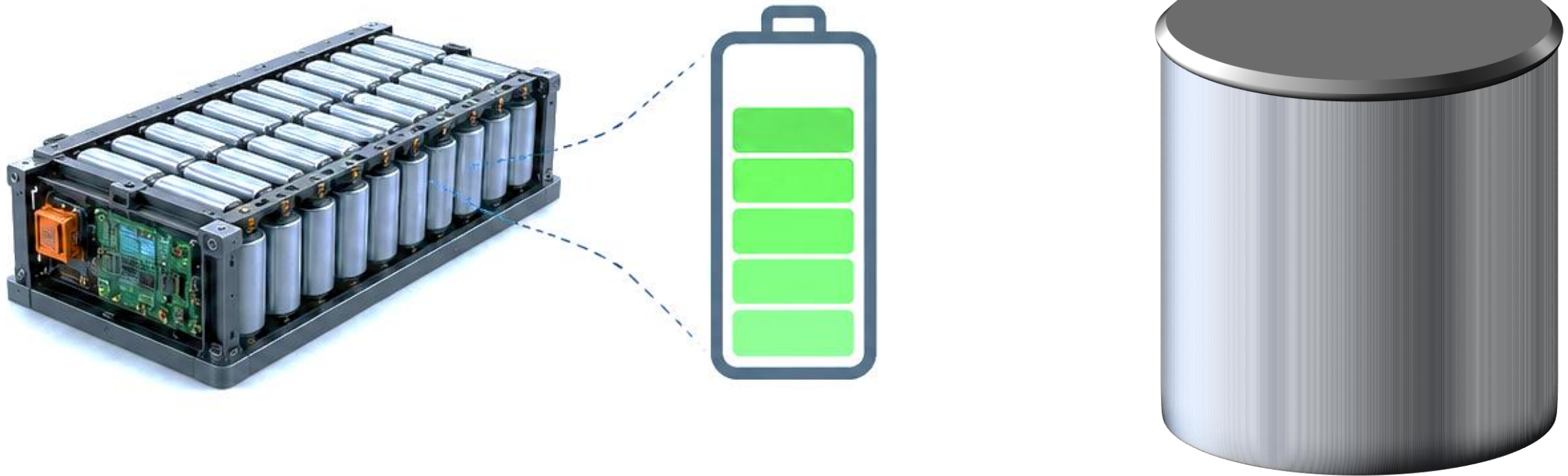
State Of Charge (SOC): The percentage of energy remaining in a battery relative to its full capacity.



① State Of Charge



State Of Charge (SOC): The percentage of energy remaining in a battery relative to its full capacity.



① State Of Charge

Importance of State of Charge (SOC) in Electric Vehicles:

- Indicates the remaining battery charge level
- Estimates the remaining driving range
- Determines when recharging is required
- Supports efficient energy management
- Prevents overcharging and deep discharging
- Enhances battery lifespan and safety
- Provides critical information for Battery Management Systems (BMS)





① State Of Charge

SOC cannot be accurately estimated from voltage alone due to the nonlinear relationship between voltage and state of charge!

① State Of Charge

SOC cannot be accurately estimated from voltage alone due to the nonlinear relationship between voltage and state of charge!



Maybe he has the flu.

Maybe he has warmed up after exercise.

Maybe he has been in a hot environment.

-
-
-



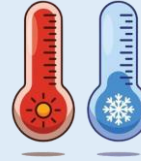
① State Of Charge

SOC cannot be accurately estimated from voltage alone due to the nonlinear relationship between voltage and state of charge!

State	Condition	Voltage
50%	Charge	3.75 V
50%	Discharge	3.65 V

① State Of Charge

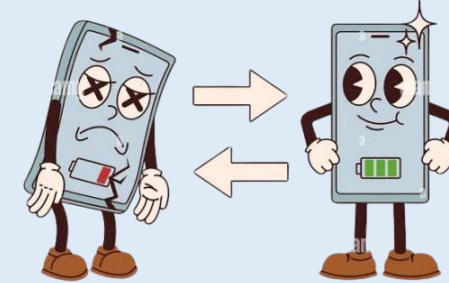
Battery Temperature



Environment Temperature



Battery Aging (SOH)



Load/Power Demand



Current/Voltage

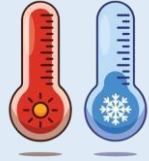


SOC

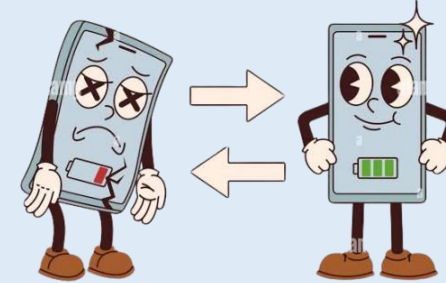
State of Charge

① State Of Charge

Battery Temperature



Battery Aging (SOH)



Environment Temperature



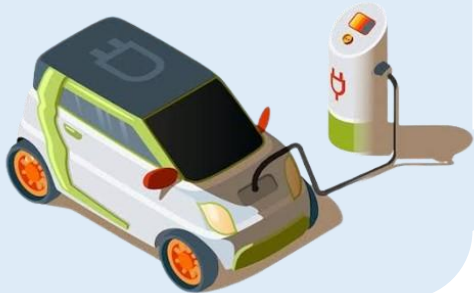
Load/Power Demand



SOC

State of Charge

Current/Voltage



SOC is the result of multiple interacting factors, not just the measured voltage!

① State Of Charge

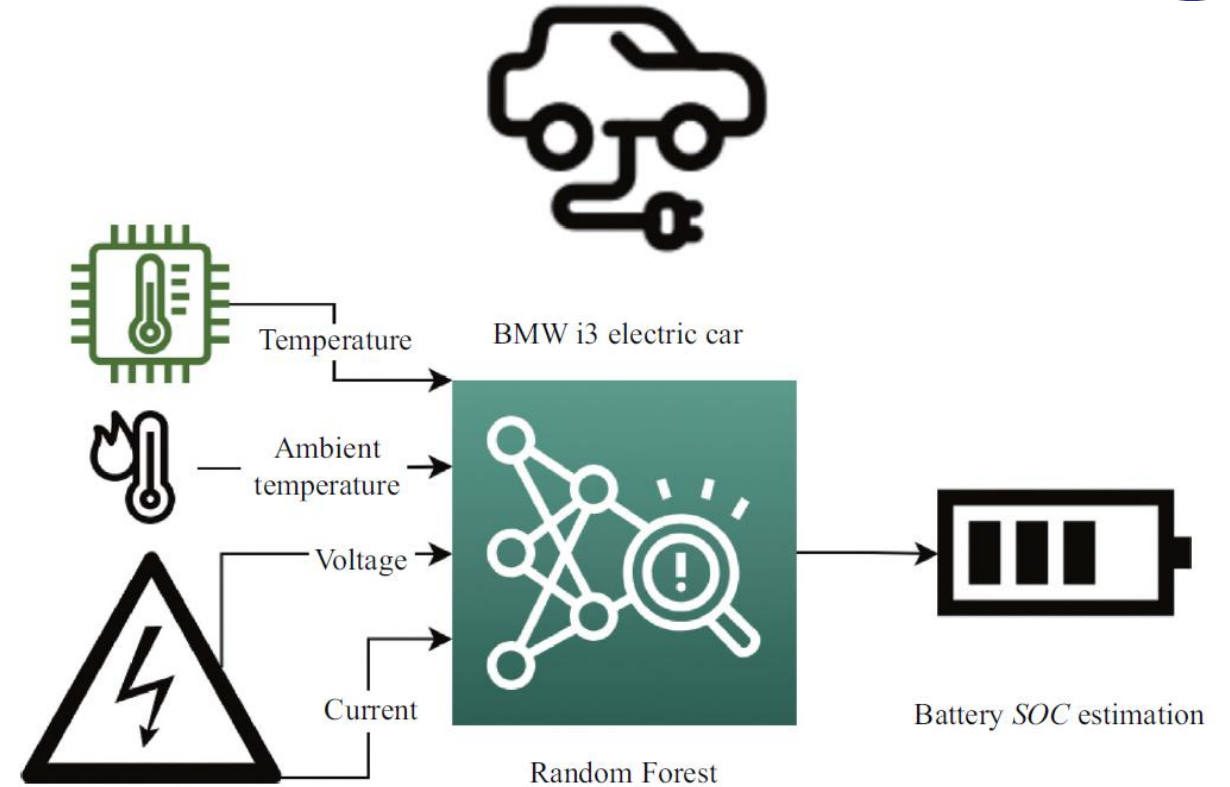
Machine Learning



SOC is the result of multiple interacting factors, not just the measured voltage!

① State Of Charge

Machine Learning



SOC is the result of multiple interacting factors, not just the measured voltage!



① State Of Charge

1	Data collection and preprocessing
2	Data splitting
3	Model configuration
4	k-fold cross-validation
5	Hyperparameter tuning - number of trees
6	Model training
7	Model evaluation
8	Results and analysis

① State Of Charge

1 Data collection and preprocessing

Gather real-world driving trip profiles and record relevant data:
Voltage (V), current (A), battery temperature (°C), and ambient temperature (°C).
Combine the data into a single dataset, ensuring proper alignment of corresponding readings.

In this research, the simulation experiments will make use of an actual dataset comprising **70 journeys** made by a **BMW i3** electric vehicle, which is equipped with a 60 Ah battery pack.

This data has been gathered using **electric vehicle sensors** installed on the car and is sampled at a 1Hz rate through the OBD port.



The dataset used in this study is openly accessible on IEEE Data port at the following link:
<https://ieee-dataport.org/open-access/battery-and-heating-data-real-driving-cycles>.



① State Of Charge

1

Data collection and preprocessing

```
Voltage = [360, 358, None, 352]
Current = [25, 28, 30, 35]
BatteryTemp = [30, 31, 32, 33]
AmbientTemp = [20, 20, 21, 22]
SOC = [80, 78, 75, 72]

#DataCleaning
clean_voltage = []

for v in Voltage:
    if v != None:
        clean_voltage.append(v)

print(clean_voltage)
```

① State Of Charge

1

Data collection and preprocessing

```
import pandas as pd  
data = pd.read_csv("battery_data.csv")  
print(data.head())
```

SOC	AmbientTemp	BatteryTemp	Current	Voltage
80	20	30	25	360
78	20	31	28	358
75	21	32	30	None
72	22	33	35	352



① State Of Charge

1

Data collection and preprocessing

```
import pandas as pd

data = pd.read_csv("battery_data.csv")
print(data.head())

#Features
X = data[['Voltage', 'Current', 'BatteryTemp', 'AmbientTemp']]

#Traget
y = data['SOC']

#DataCleaning
data = data.dropna()

OR

data.fillna(data.mean(), inplace=True)
```



① State Of Charge

2

Data splitting

Divide the dataset into training and testing sets. For instance, trips #1 to #60 can be used for training, consisting of 945,027 instances.

Trips #61 to #70 serve as the testing dataset with 118,974 instances.



① State Of Charge

2

Data splitting

```
data = [  
    [360, 25, 30, 20, 80],  
    [358, 28, 31, 20, 78],  
    [355, 30, 32, 21, 75],  
    [352, 35, 33, 22, 72]  
]  
  
# DataSplitting  
X_train = X[:3]  
X_test = X[3:]  
  
y_train = y[:3]  
y_test = y[3:]  
  
print("Training Data:", X_train)  
print("Testing Data:", X_test)
```



① State Of Charge

3

Model configuration

Configure the Random Forest model, specifying hyperparameters like the number of trees in the forest, maximum tree depth, and the number of features considered at each split. These hyperparameters can be tuned to optimize performance.



① State Of Charge

3

Model configuration

Configure the Random Forest model, specifying hyperparameters like the number of trees in the forest, maximum tree depth, and the number of features considered at each split. These hyperparameters can be tuned to optimize performance.

```
from sklearn.ensemble import RandomForestRegressor  
  
rf = RandomForestRegressor(  
    n_estimators=25,  
    random_state=42  
)
```



① State Of Charge

4

k-fold cross-validation

Implement 5-fold cross-validation ($k=5$) on the training dataset. This process involves splitting the training data into five subsets or "folds".
Train the RF model on four of these folds and validate it on the remaining one in a rotating fashion, resulting in five sets of model evaluations.



① State Of Charge

4

k-fold cross-validation

```
kf = KFold(n_splits=5, shuffle=True, random_state=42)

#Cross-validation (MSE is negative in sklearn, so we invert it)
scores = cross_val_score(
    model,
    X,
    y,
    cv=kf,
    scoring='neg_mean_squared_error'
)

#Convert to positive MSE
mse_scores = -scores

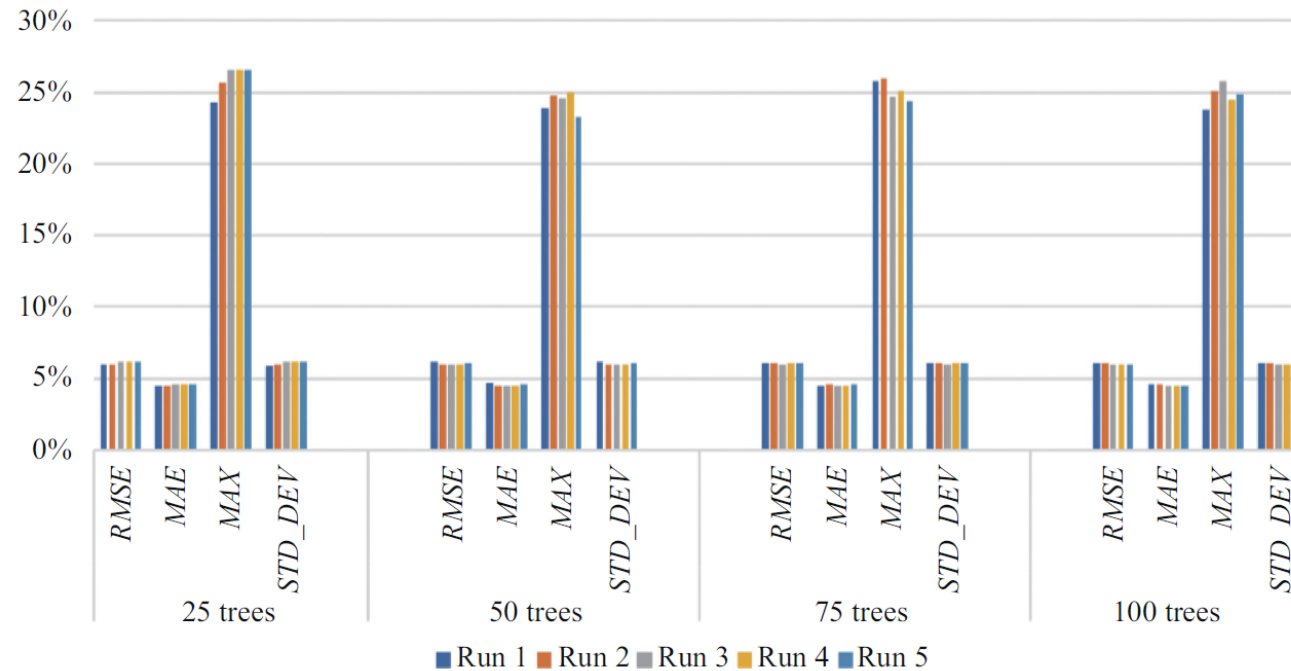
print ("MSE for each fold:", mse_scores)
print ("Mean MSE:", np.mean(mse_scores))
```

① State Of Charge

5

Hyperparameter tuning - number of trees

Conduct a hyperparameter tuning experiment to determine the optimal number of trees for the Random Forest. Trial different values such as 25 trees, 50 trees, 75 trees, and 100 trees. For each configuration, perform 5-fold cross-validation on the training data and evaluate the model's performance using metrics.



Performance metrics for various configuration of trees in RF model for SOC estimation



① State Of Charge

5

Hyperparameter tuning - number of trees

```
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import cross_val_score
import numpy as np

n_trees = [25, 50, 75, 100]

for n in n_trees:
    model = RandomForestRegressor (n_estimators=n, random_state=42)
    scores = cross_val_score (model, X, y, cv=5, scoring='neg_mean_squared_error')
    print (n, "trees -> MSE:", -np.mean(scores))
```



① State Of Charge

6

Model training

Based on the hyperparameter tuning results, select the optimal number of trees that yields the best performance on the training dataset.



① State Of Charge

7

Model evaluation

Employ the trained RF model with the chosen number of trees to predict SOC values for the testing dataset. Calculate evaluation metrics such as RMSE, MAE, MAX, and standard deviations to assess the model's performance.



① State Of Charge

7

Model evaluation

```
from sklearn.model_selection import cross_val_score, KFold
from sklearn.ensemble import RandomForestRegressor
import numpy as np

model = RandomForestRegressor(n_estimators=25, random_state=42)
kf = KFold(n_splits=5, shuffle=True, random_state=42)

mae = -cross_val_score(model, X, y, cv=kf, scoring='neg_mean_absolute_error')
mse = -cross_val_score(model, X, y, cv=kf, scoring='neg_mean_squared_error')
r2 = cross_val_score(model, X, y, cv=kf, scoring='r2')

print ("MAE:", np.mean(mae))
print ("MSE:", np.mean(mse))
print ("RMSE:", np.sqrt(np.mean(mse)))
print ("R2 Score:", np.mean(r2))
```



① State Of Charge

8

Results and analysis

Analyze the results of the RF-based SOC estimation, emphasizing the accuracy, robustness, and consistency achieved with the chosen number of trees.

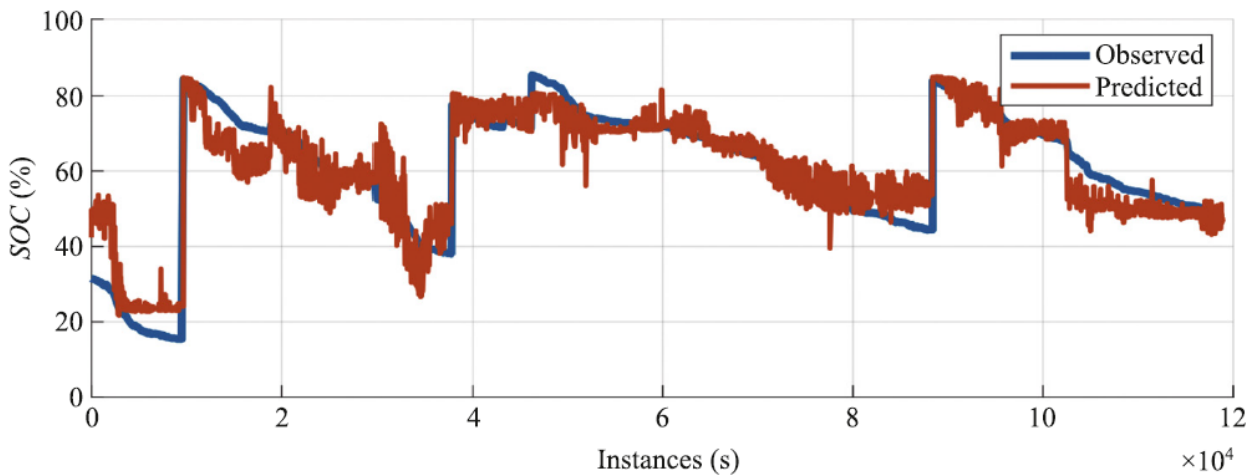
```
import matplotlib.pyplot as plt
plt.plot(y_test.values, label='Real SOC')
plt.plot(y_pred, label='Predicted SOC')
plt.xlabel('Instances')
plt.ylabel('SOC (%)')
plt.legend()
plt.show()
```

① State Of Charge

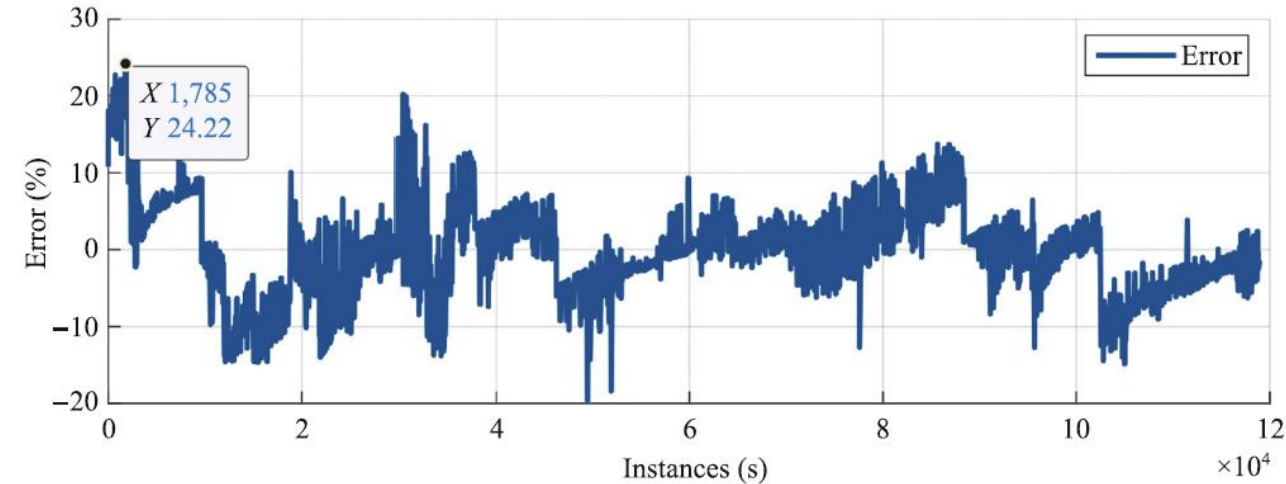
8

Results and analysis

These results highlight the superior predictive capability of RF over ELM in the SOC estimation task, making RF the more effective and precise approach in this study.



Results of the SOC estimation by RF for the testing data.



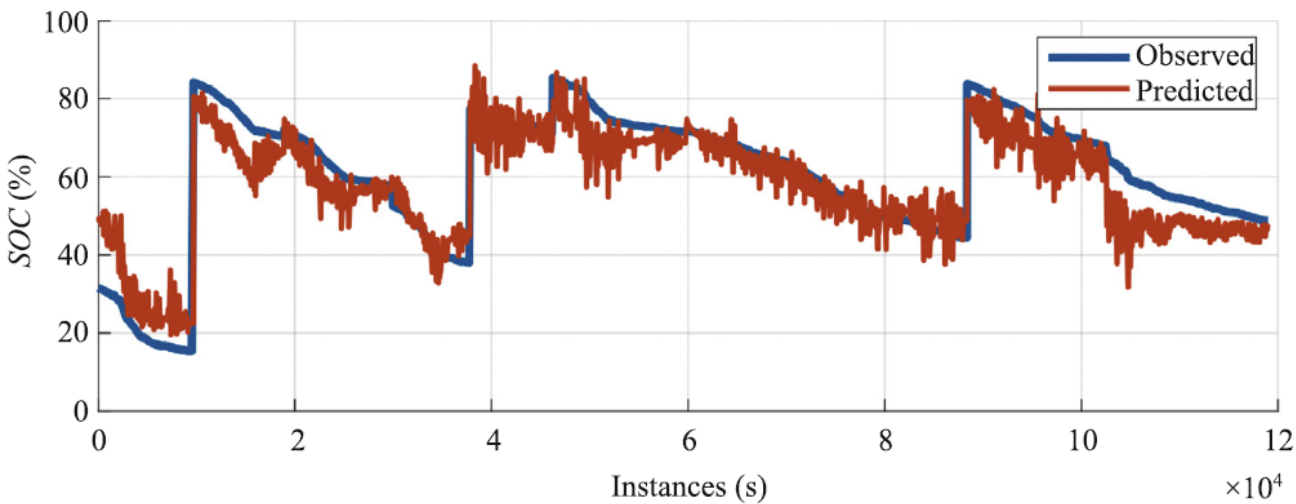
Results of the error between actual and predicted.

① State Of Charge

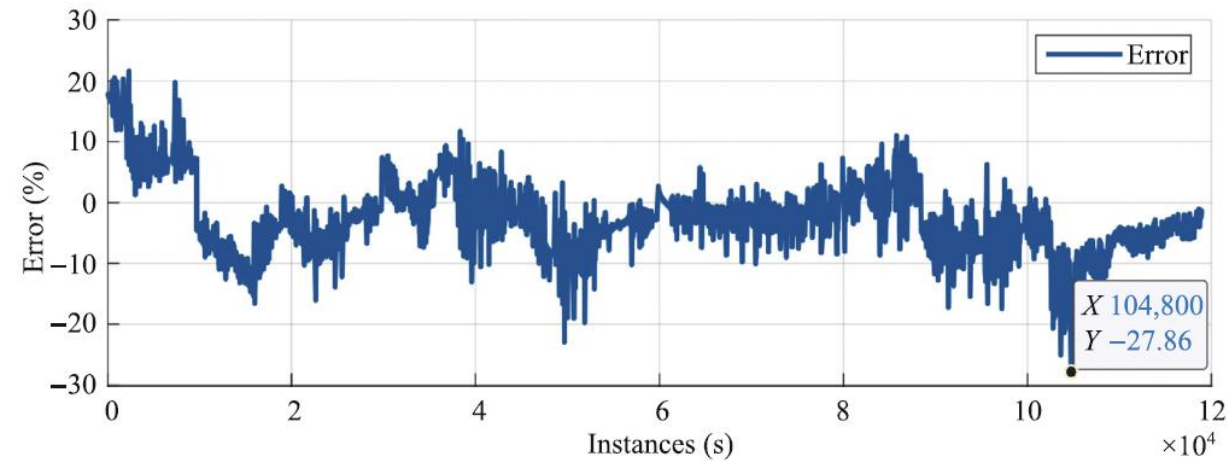
8

Results and analysis

These results highlight the superior predictive capability of RF over ELM in the SOC estimation task, making RF the more effective and precise approach in this study.



Results of the SOC estimation by ELM for the testing data.



Results of the error between actual and predicted.



① State Of Charge

8

Results and analysis

These results highlight the superior predictive capability of RF over ELM in the SOC estimation task, making RF the more effective and precise approach in this study.

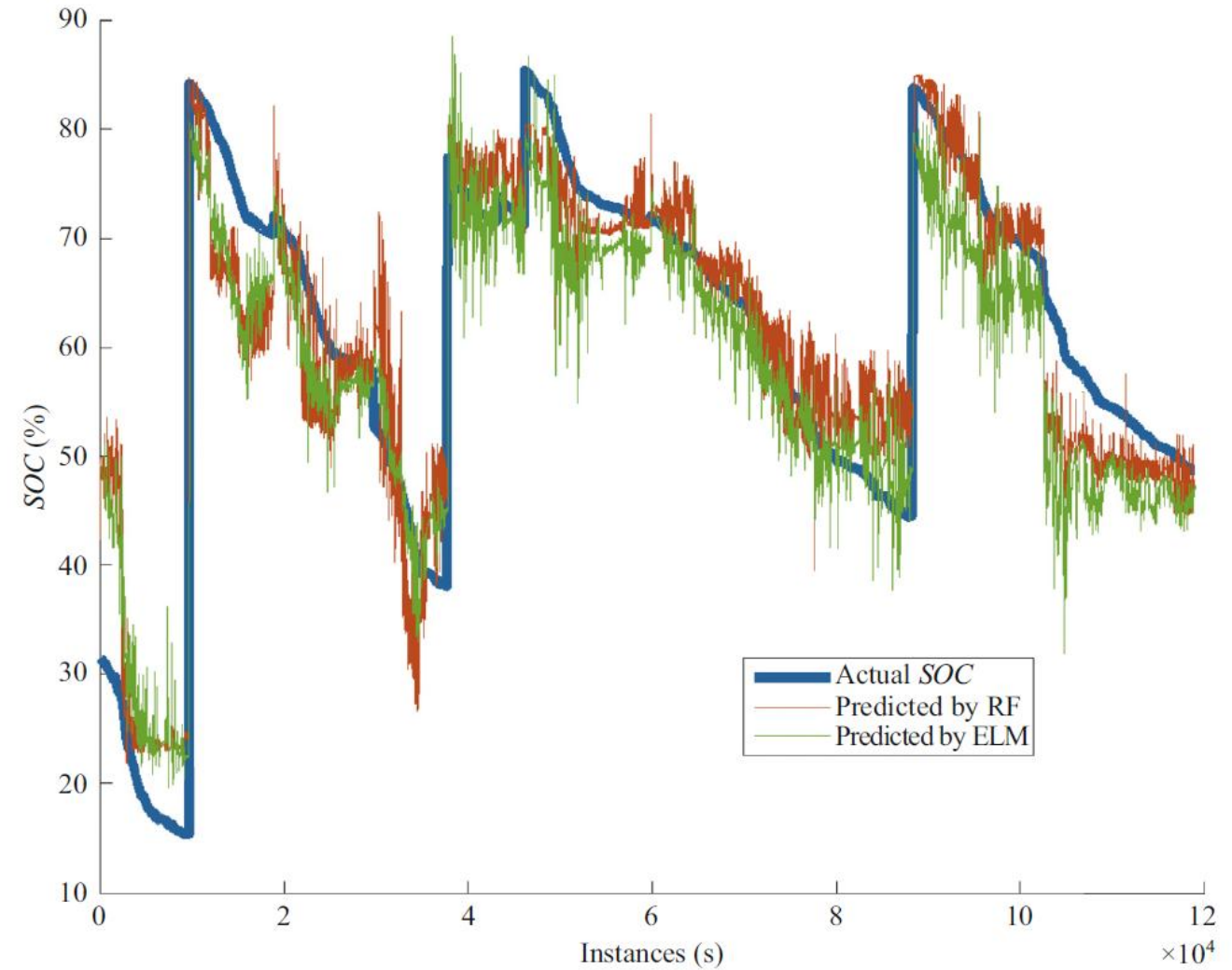
Performances of RF with ELM.

Performance	Best					Average				
Evaluation (%)	RF (25 trees)	RF (50 trees)	RF (75 trees)	RF (100 trees)	ELM	RF (25 trees)	RF (50 trees)	RF (75 trees)	RF (100 trees)	ELM
RMSE	5.902,8	5.925,6	5.994,9	5.954,7	6.312,7	6.062,2	5.999,2	6.022,6	5.989,5	6.564,7
MAE	4.432,1	4.415,2	4.478,8	4.476,7	5.111,2	4.512,9	4.510,6	4.509,0	4.489,4	4.986,6
MAX	24.217,5	24.556,8	24.693,8	25.795,8	27.859,7	25.924,5	24.299,2	25.159,6	24.786,9	28.190,7
STD. DEV.	5.899,9	5.923,1	5.994,8	5.954,5	5.673,9	6.061,6	5.998,2	6.021,6	5.989,2	6.105,9
Computation time (s)	165.28	347.86	526.50	656.11	1.42					

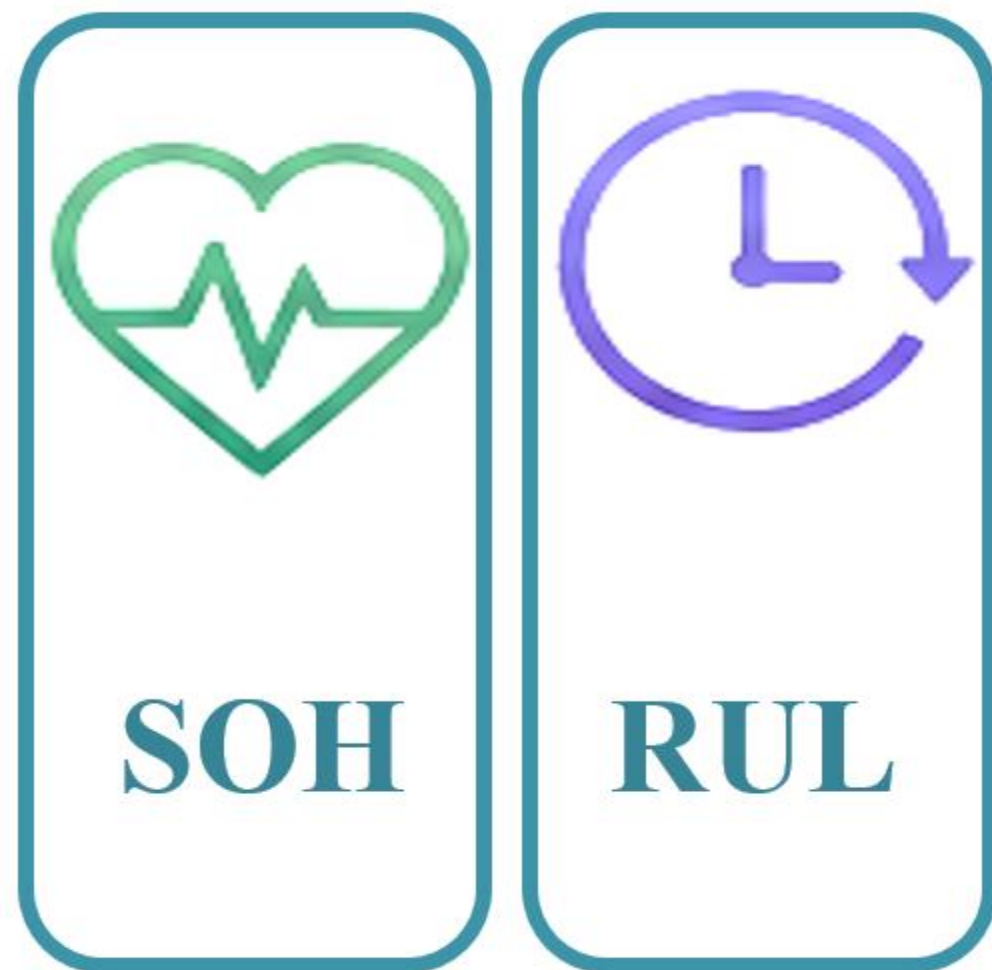
① State Of Charge

8

Results and analysis



Comparison performances of RF and ELM for SOC estimation.



②



The Challenges of PHM (Prognostics and Health Management) in Electric Vehicles

Challenge	Why It's Difficult
Complexity	Batteries are multiphysics (heat, electricity, chemistry) and multiscale (from microscopic cell particles to the whole pack).
Nonlinearity	Aging does not follow a simple linear path, making it hard to predict the "knee point" where performance drops sharply.
Data Limitations	Real-world field data is often "noisy," incomplete, and captured under uncontrolled, dynamic driving conditions.
Variability	Every battery cell is slightly different (cell-to-cell variation), and each vehicle experiences different usage patterns and environments.
Computational Costs	Accurate models require significant power, which is restricted within a vehicle's onboard Battery Management System (BMS).



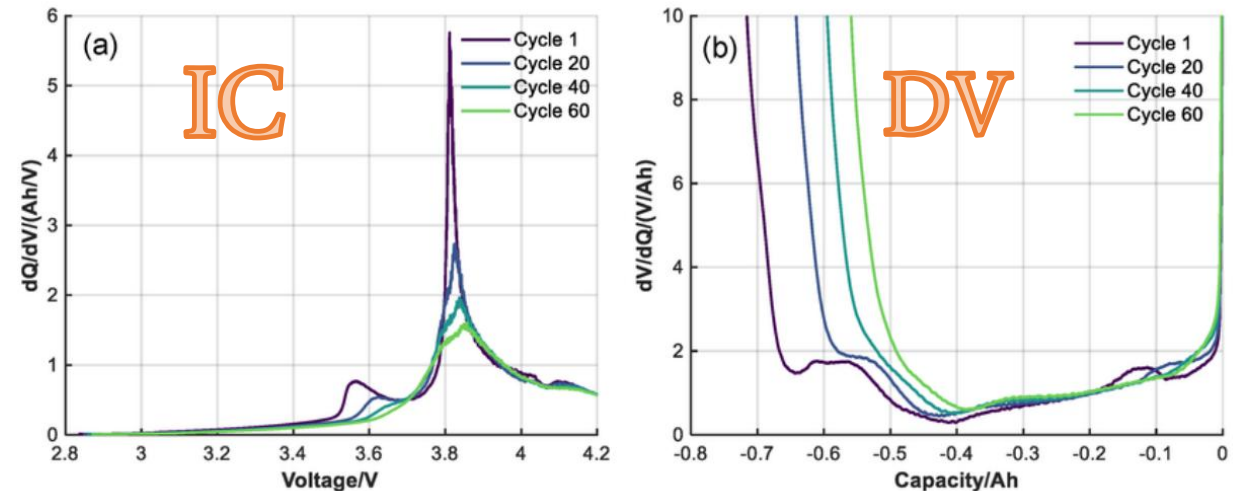
These techniques are critical because they bridge the gap between "black-box" AI models and the physical reality of the battery.

Incremental Capacity (IC) Analysis($\frac{\partial Q}{\partial V}$):

- If you plot IC you get a curve with peaks and valleys. Each peak represents a specific electrochemical process. As the battery ages, these peaks will shift (move left or right), shrink (lose height), or broaden (widen). These changes are direct indicators of specific chemical failures.

Differential Voltage (DV) Analysis($\frac{\partial V}{\partial Q}$):

- It often produces a curve where degradation is marked by the movement of inflection points. It is particularly sensitive to anode-side degradation, making it a perfect partner to IC analysis, which is often more sensitive to the cathode





Key Challenges in Battery State Of Health(SOH) Prediction

Spatiotemporal Heterogeneity: Data quality and availability vary significantly over time and across different vehicle fleets

Operational Variability: Real-world conditions involve unpredictable voltage windows, inconsistent charging/discharging currents, and fluctuating environmental factors like temperature and humidity

Knowledge Gaps: There is a lack of precise, universally accepted knowledge regarding specific battery fade mechanisms and the inherent randomness in data.

Reproducibility: Laboratory experiments are costly, time-consuming, and often fail to capture the complexity of "in situ" (real-world) usage, making the transfer of academic models to online/cloud-based prognostics difficult.

Choosing Approach To Solving These Problems

② SOH & RUL

Datasets in this study.

Datasets	Quantity and Sources	Time span	Research
Group A (Cell level)	420 large format LiFePO ₄ cells are collected from different vehicles (300 in the training set; 120 in the testing set)	Different phases of capacity fade cycled under realistic vehicle conditions.	Estimation of SOH for battery cells
Group B (Pack level)	9 EVs, consisting of 1602 cells (178 cells are connected in series for each vehicle)	8 consecutive months of different service time of EVs, with the driving distance from 40,000 to 200,000 km, service time from 218 to 468 days.	Prediction of battery RUL in terms of capacity under the different conditions of driving distance and service life for EVs



② SOH & RUL

Feature engineering

Because EV field data is notoriously noisy, directly differentiating the capacity-voltage curves leads to massive spikes in the data. The study uses a two-step process to fix this before any features are extracted.

Step A: Trajectory-based Piecewise Linear / Reduced-order Modeling

Instead of using a single, highly complex nonlinear equation for the battery, the method breaks the nonlinear system down into simpler, linear regions.

A general nonlinear system is written as $\frac{dx(t)}{dt} = f(x(t)) + Bu(t)$ and $y(t) = C^T x(t)$.

Linearized models are generated around specific states $(x_0, x_1, \dots, x_{s-1})$ using Jacobian matrices. For example, the system at state x_i becomes $\frac{dx(t)}{dt} = f(x_i) + A_i(x - x_i) + B_i u$, where A_i is the Jacobian matrix.

To further simplify computation, the model uses **Krylov projection** to reduce the dimensionality of the state space, represented as a projection basis $V \in \mathbb{R}^{n \times q}$ where $x = Vz$. The final curve is a combination of these linear segments blended together with specific weights (w_i).

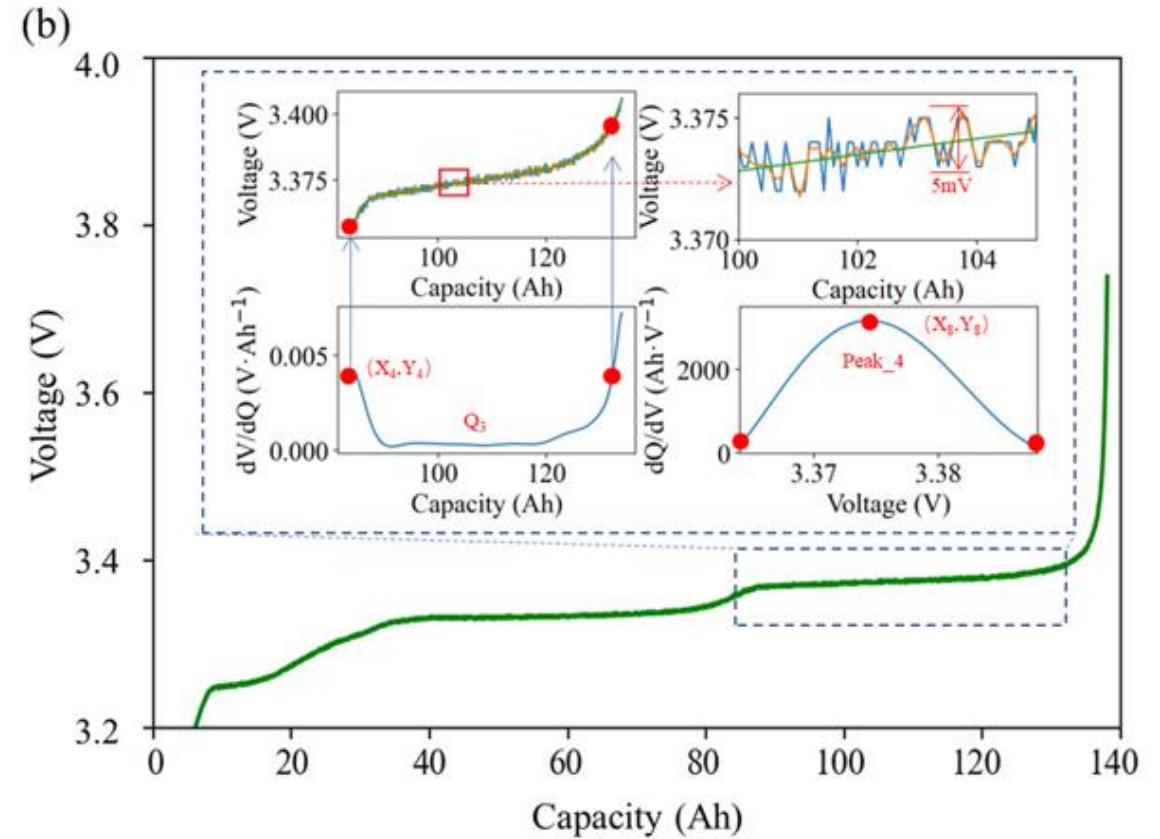
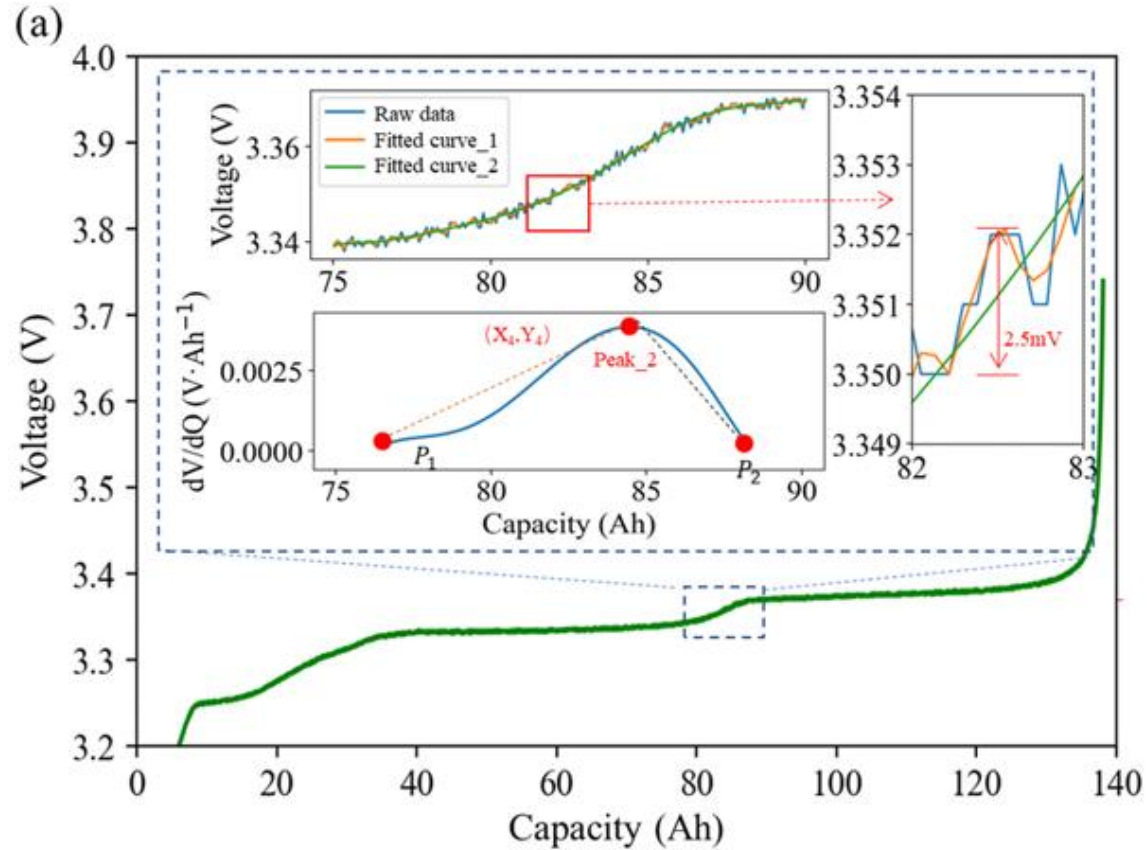
Step B: Exponentially Weighted Moving Average (EWMA)

After the mathematical curve fitting, an EWMA is applied. Unlike a Simple Moving Average (SMA) which treats all historical data equally, EWMA prioritizes recent data to smooth out time-series trends without losing the immediate signal shape.

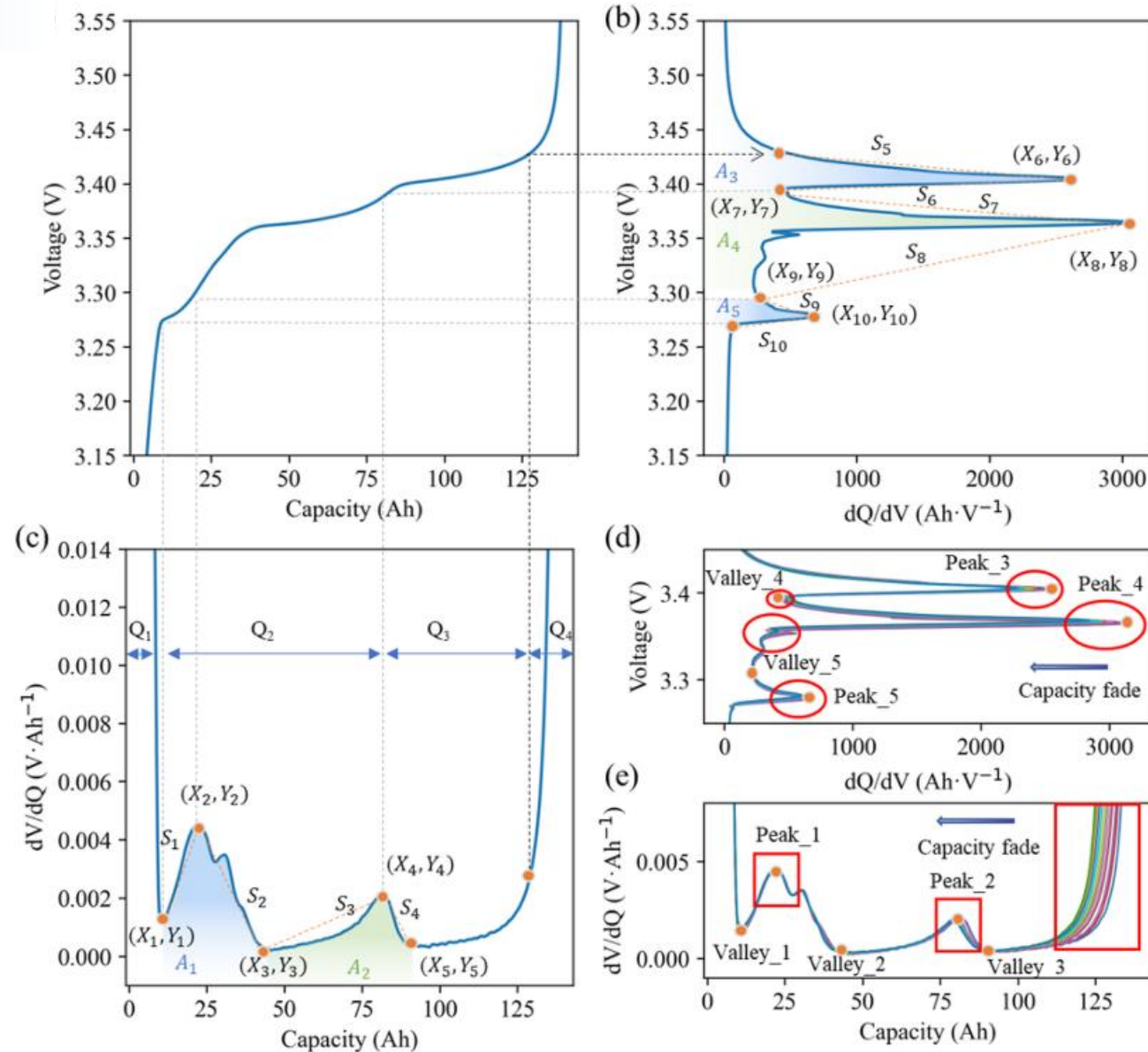
The recursive formula is $EWMA_t = \alpha \times r_t + (1 - \alpha) \times EWMA_{t-1}$.

The coefficient α controls the weighting and is determined in practice by $\alpha = \frac{2}{n+1}$

② SOH & RUL



② SOH & RUL



② SOH & RUL

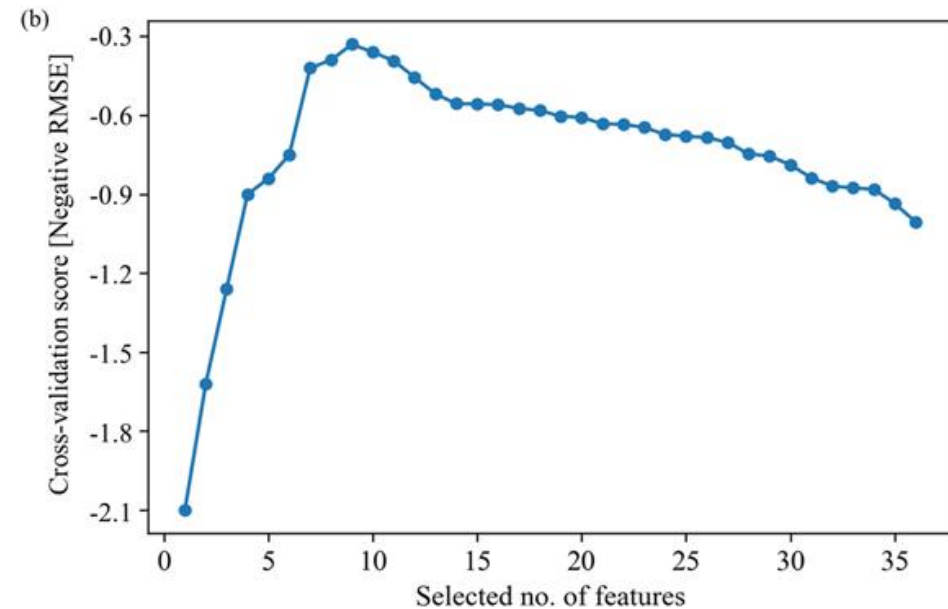
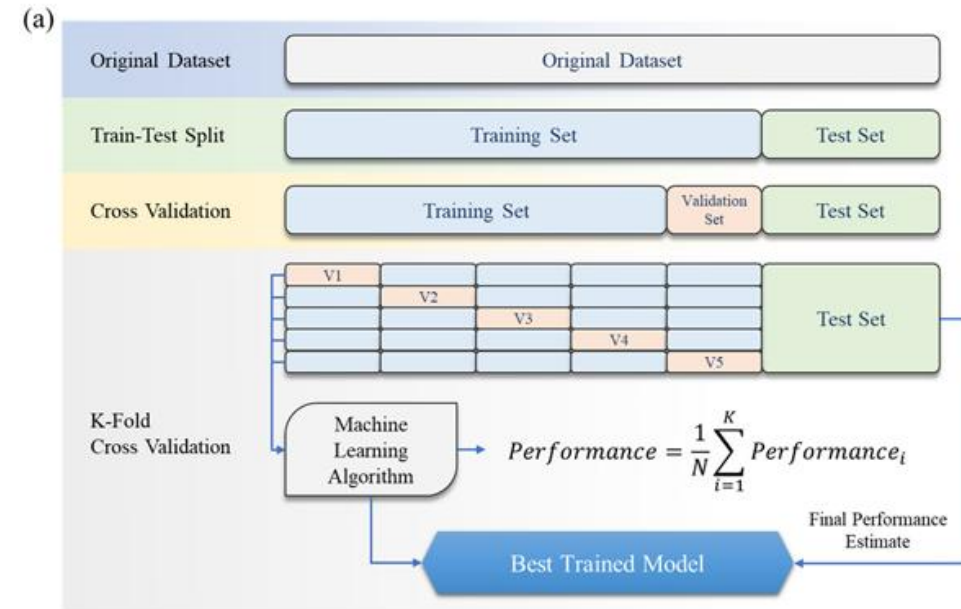
DV analysis		IC Analysis	
(1)	Cumulated Charge Capacity [Ah]: Q_1	(1)	The x-Coordinate of the Peak_3: X_6
(2)	Cumulated Charge Capacity [Ah]: Q_2	(2)	The y-Coordinate of the Peak_3: Y_6
(3)	Cumulated Charge Capacity [Ah]: Q_3^*	(3)	The x-Coordinate of the Valley_4: X_7
(4)	Cumulated Charge Capacity [Ah]: Q_4	(4)	The y-Coordinate of the Valley_4: Y_7
(5)	The x-Coordinate of the Valley_1: X_1	(5)	The x-Coordinate of the Peak_4: X_8^*
(6)	The y-Coordinate of the Valley_1: Y_1	(6)	The y-Coordinate of the Peak_4: Y_8^*
(7)	The x-Coordinate of the Peak_1: X_2	(7)	The x-Coordinate of the Valley_5: X_9
(8)	The y-Coordinate of the Peak_1: Y_2	(8)	The y-Coordinate of the Valley_5: Y_9
(9)	The x-Coordinate of the Valley_2: X_3	(9)	The x-Coordinate of the Peak_5: X_{10}
(10)	The y-Coordinate of the Valley_2: Y_3	(10)	The y-Coordinate of the Peak_5: Y_{10}
(11)	The x-Coordinate of the Peak_2: X_4	(11)	Slope of the segment of the curve: S_5
(12)	The y-Coordinate of the Peak_2: Y_4^*	(12)	Slope of the segment of the curve: S_6
(13)	The x-Coordinate of the Valley_3: X_5	(13)	Slope of the segment of the curve: S_7^*
(14)	The y-Coordinate of the Valley_3: Y_5	(14)	Slope of the segment of the curve: S_8^*
(15)	Slope of the segment of the curve: S_1	(15)	Slope of the segment of the curve: S_9
(16)	Slope of the segment of the curve: S_2	(16)	Slope of the segment of the curve: S_{10}
(17)	Slope of the segment of the curve: S_3^*	(17)	Area between cut-off voltage and Valley_4: A_3
(18)	Slope of the segment of the curve: S_4^*	(18)	Area between Valley_4 and Valley_5: A_4^*
(19)	Area between Valley_1 and Valley_2: A_1	(19)	Area between Valley_5 and the inflection point: A_5
(20)	Area between Valley_2 and Valley_3: A_2		



② SOH & RUL

Feature Selection Strategy

- Objective:** Identify the most significant subset of input features to reduce computational complexity, accelerate training, and improve model generalization.
- Methodology:** Recursive Feature Elimination with Cross-Validation (RFE-CV).
- Algorithm:** Random Forest (RF) utilizing 500 decision trees to rank feature significance.
- Process:** Employs 10-fold cross-validation to iteratively remove uninformative features and re-rank the remaining ones, aiming to minimize Root-Mean-Square Error (RMSE).
- Outcome:** The pipeline successfully engineered and isolated the 9 most critical features yielding the lowest RMSE.





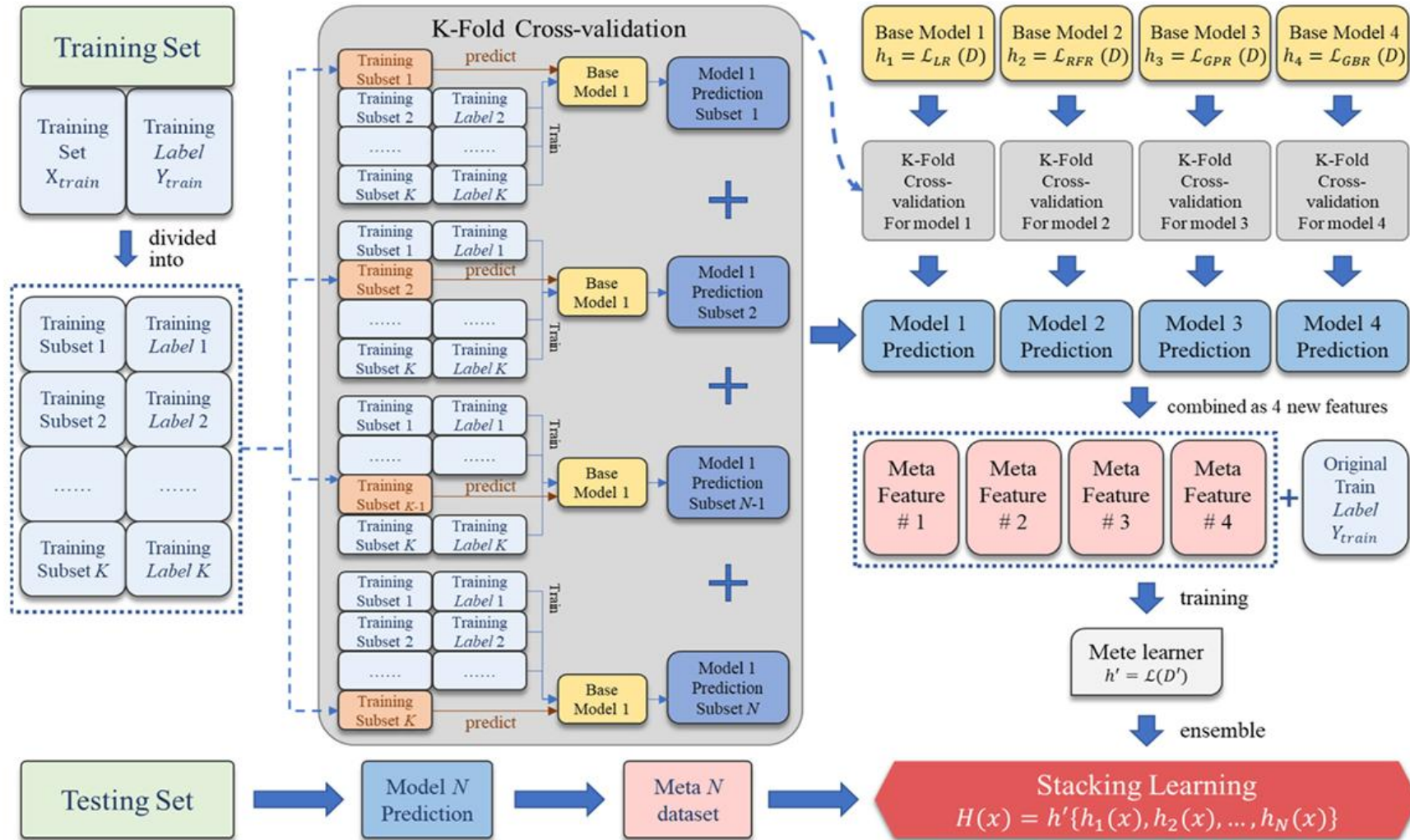
② SOH & RUL

The Stacking Ensemble Strategy

Instead of relying on a single model (which may have high bias or poor generalization), the stacking approach creates a two-layer hierarchy:

- **Layer 1 (Base Learners):** Four distinct, heterogeneous models are trained in parallel using the same 9 engineered features. Each model interprets the data differently:
- **Linear Regression (LR):** Provides physical interpretability but struggles with non-linear complexities.
- **Random Forest Regressor (RFR):** Excellent for high-dimensional data and capturing non-linear relationships via decision trees.
- **Gaussian Process Regression (GPR):** Highly effective at quantifying uncertainty and handling noisy data using probabilistic modeling.
- **Gradient Boosting Regressor (GBR):** A sequential ensemble that focuses on correcting errors from previous trees to build a "strong" predictor from "weak" learners.
- **Layer 2 (Meta-Learner):** A final model (Linear Regression) takes the outputs of the four base models as its input. Its job is to "learn" how to best weight or combine those predictions to produce the final, most accurate battery State of Health (SOH) estimate.

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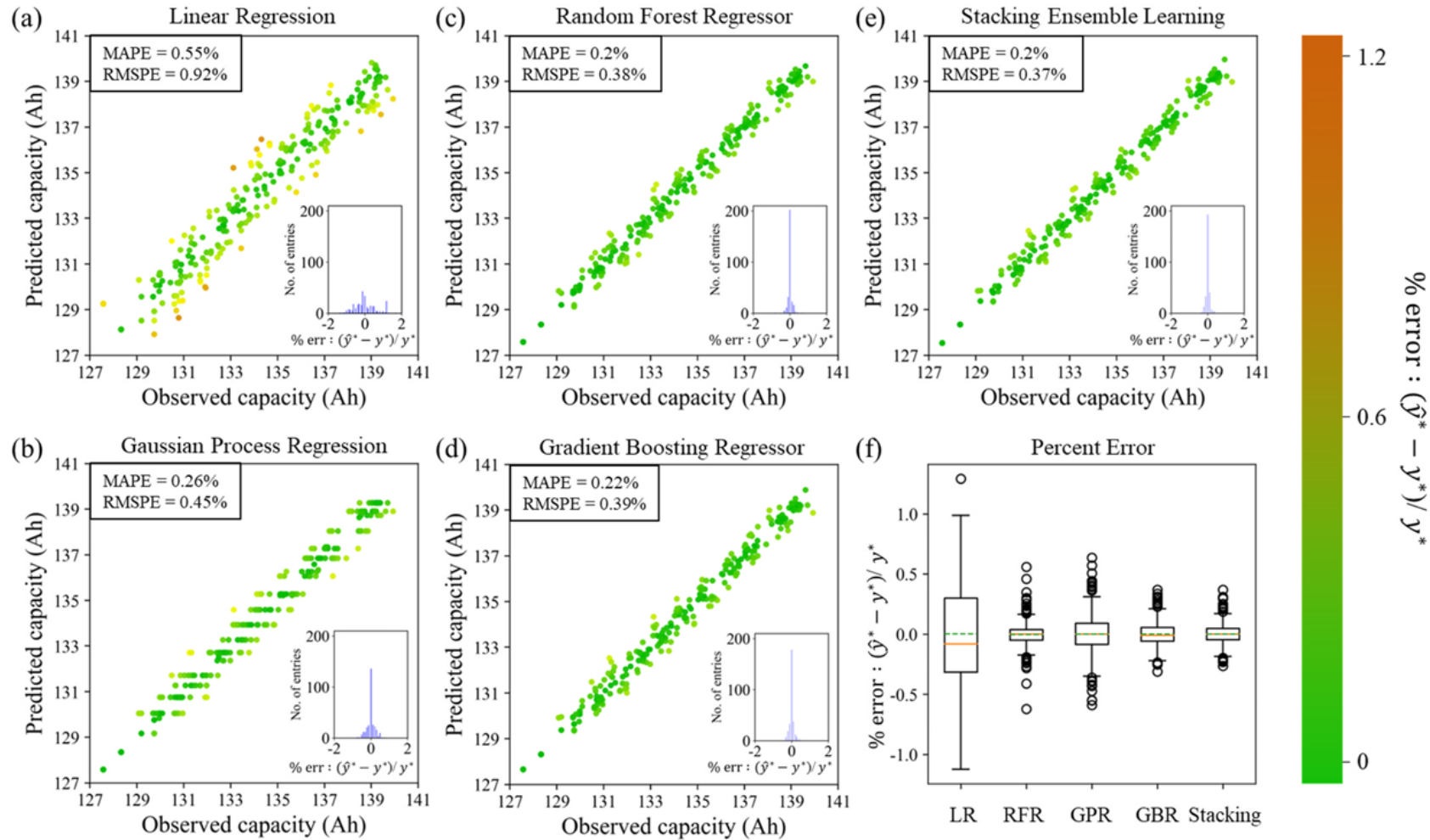
Performance Metrics

To quantitatively measure how closely the predicted capacity matches the true capacity, three specific metrics are used:

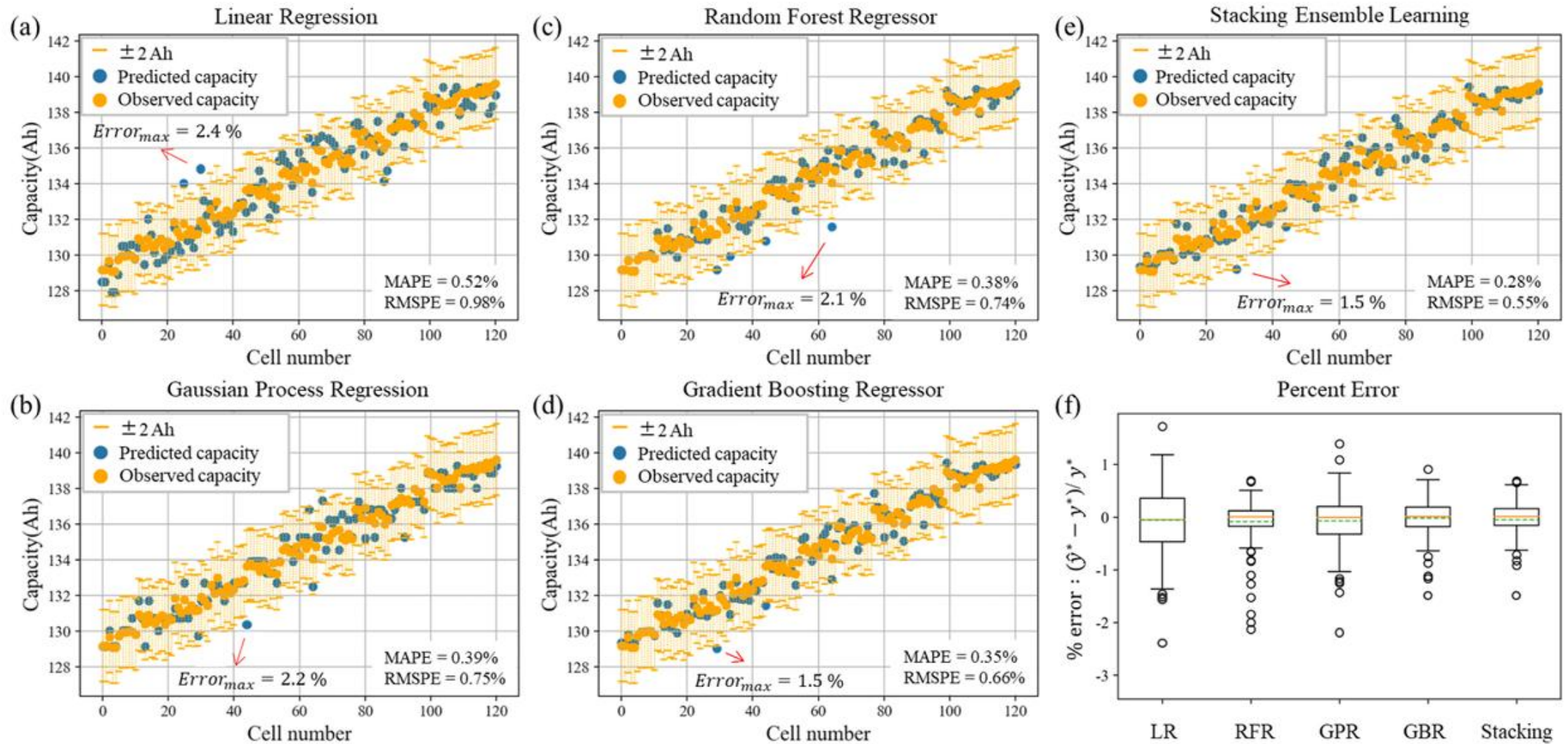
Metric	Full Name	Purpose
MAPE	Mean Absolute Percentage Error	Measures the average percentage deviation of the model.
RMSPE	Root Mean Square Percentage Error	Highlights larger errors (more sensitive to outliers).
Error_max	Maximum Absolute Error	Defines the "worst-case" scenario for safety and reliability.

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Estimation for the battery SOH at the cell level



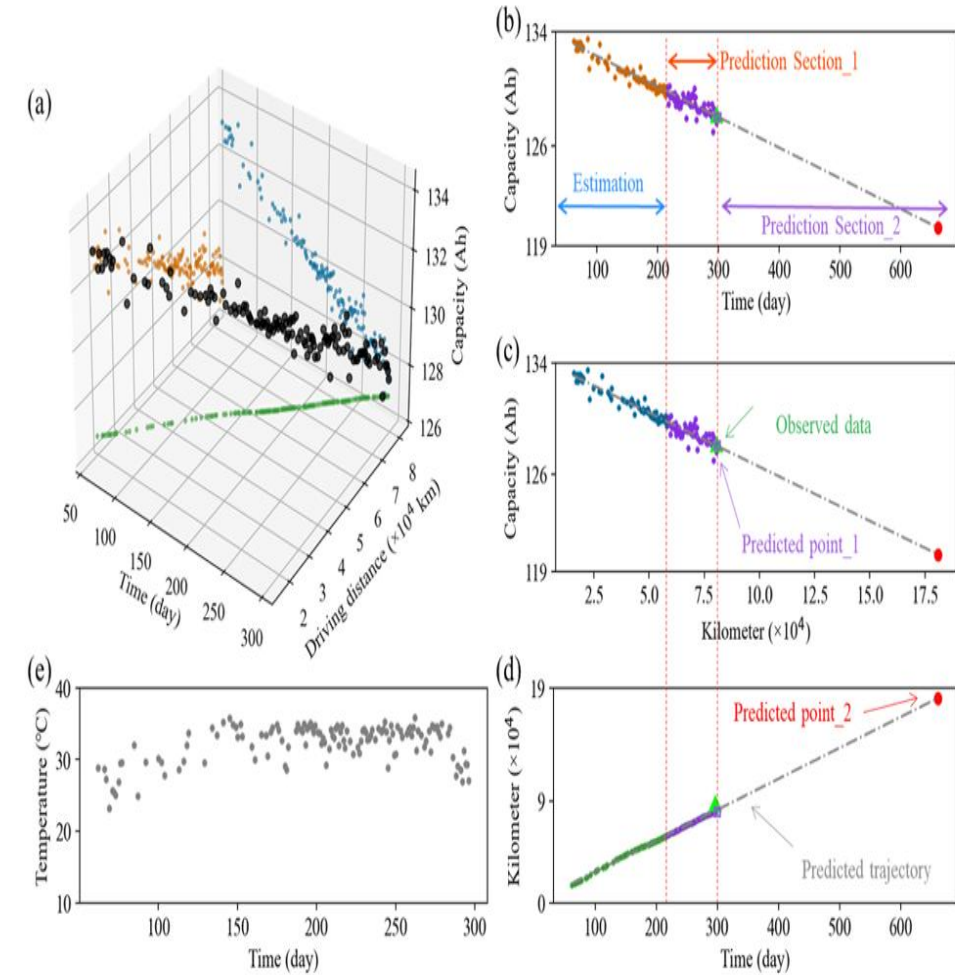
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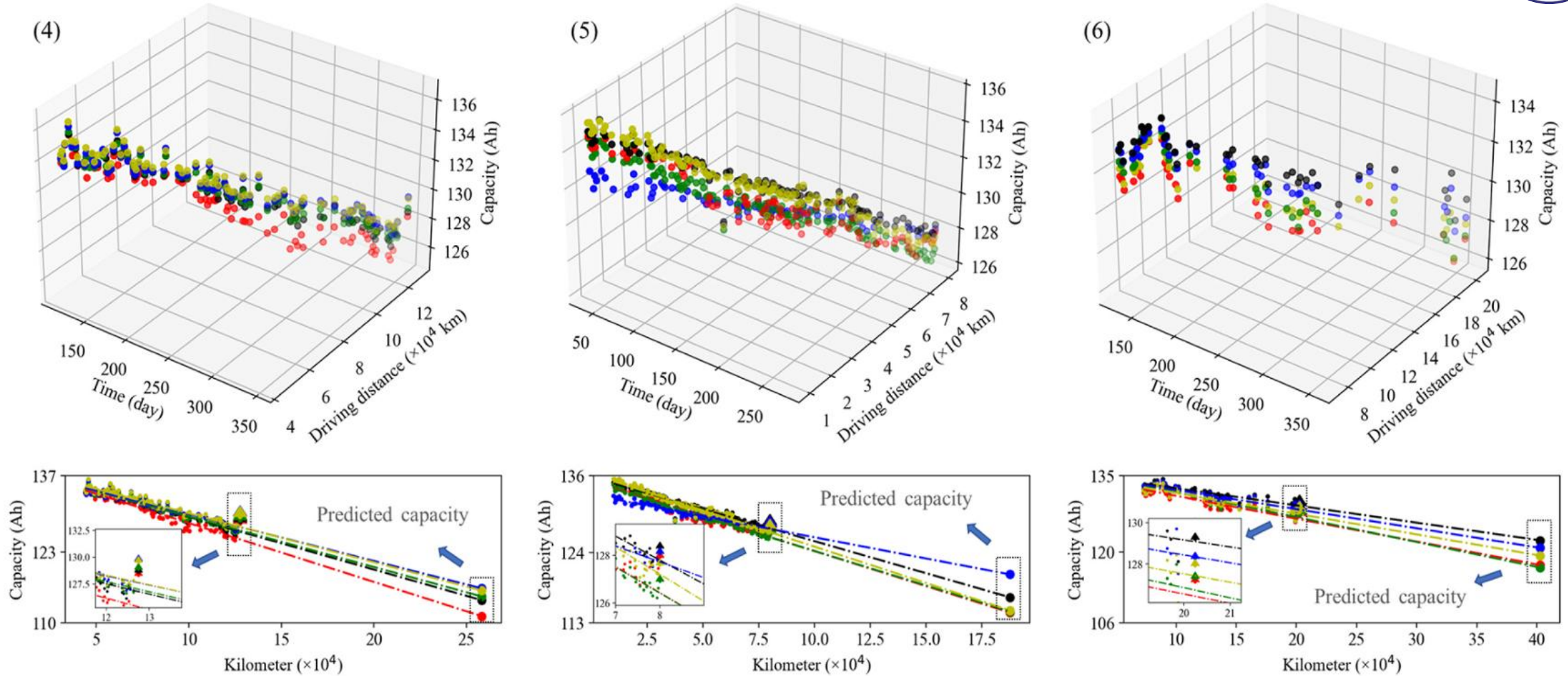
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RUL prediction

- Summary:** Developed a robust Stacking Ensemble ML model using only basic onboard measurements (Voltage, Current, Temp).
- Innovation:** Overcame real-world data noise via two-stage noise reduction and physically consistent IC/DV feature extraction.
- Impact:** Enables intelligent, data-driven battery management, providing predictive health updates (RUL) directly to EV users.



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Category	Challenge	Future Solution
Charging Modes	Inability to handle Fast Charging and Partial Cycles	Develop new protocols for high-rate/non-constant current
User Behavior	Random/Arbitrary charging makes prediction difficult	Expand big-data analysis for diverse user patterns
Thermal Effects	Complex low-temperature aging mechanisms	Integrate Arrhenius-based thermal aging models
Data Complexity	Short-range cycles lack enough physical features	Implement PINNs (Physics-Informed Neural Networks)
Computation	Complex PINN loss functions are heavy for BMS	Optimize network architectures for onboard hardware

**Thanks for
your
attention!**